

EEE4114F: Digital Signal Processing

Class Test

14 March 2023

SOLUTIONS

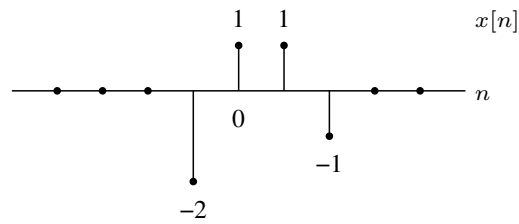
Please do not put your name or student number anywhere on this script.

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Information

- The test is closed-book.
 - This test has *four* questions, totalling 20 marks.
 - Answer *all* the questions.
 - You have 60 minutes.
 - An information sheet is attached.
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1. (5 marks) Consider the signal below:

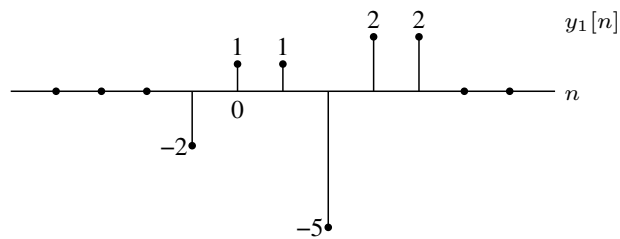


- (a) Sketch $y_1[n] = \sum_{m=-1}^1 (m+1)x[n-3m]$.
 (b) Sketch $y_2[n] = x[n] * x[-n]$.

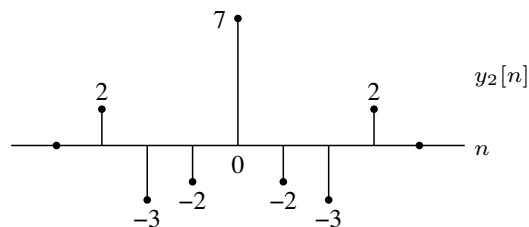
- (a) The expression can be written as

$$y_1[n] = \sum_{m=-1}^1 (m+1)x[n-3m] = 0 + x[n] + 2x[n-3],$$

and the plot is as follows:



- (b) Using graphical or other methods:



2. (5 marks) Consider a system with input $x[n]$ and output $y[n]$ that satisfies

$$y[n] = \begin{cases} x[n+1] & n > 0 \\ x[n-1] & n \leq 0. \end{cases}$$

- (a) Find the response of the system to a unit step $x_1[n] = u[n]$.
- (b) Find the response of the system to a shifted unit step $x_2[n] = u[n-1]$.
- (c) Is the system time invariant?
- (d) Prove that the system is linear.

(a) Observe that

$$y[-2] = x[-3], \quad y[-1] = x[-2], \quad y[0] = x[-1], \quad y[1] = x[2], \quad y[2] = x[3],$$

so the response to $x_1[n] = u[n]$ is seen to be $y_1[n] = u[n-1]$.

- (b) The response to $x_2[n] = u[n-1]$ can similarly be seen to be $y_2[n] = u[n-1]$.
- (c) The input $x_2[n]$ is the input $x_1[n]$ delayed by one sample, but we do not have $y_2[n] = y_1[n-1]$. Thus the system is not time invariant.
- (d) Suppose $y_1[n]$ is the response to $x_1[n]$ and $y_2[n]$ is the response to $x_2[n]$. Then

$$y_1[n] = \begin{cases} x_1[n+1] & n > 0 \\ x_1[n-1] & n \leq 0 \end{cases} \quad \text{and} \quad y_2[n] = \begin{cases} x_2[n+1] & n > 0 \\ x_2[n-1] & n \leq 0. \end{cases}$$

Now, if $y_3[n]$ the response to $x_3[n]$ then

$$y_3[n] = \begin{cases} x_3[n+1] & n > 0 \\ x_3[n-1] & n \leq 0. \end{cases}$$

Consider the case where $x_3[n] = ax_1[n] + bx_2[n]$. Then

$$y_3[n] = \begin{cases} ax_1[n+1] + bx_2[n+1] & n > 0 \\ ax_1[n-1] + bx_2[n-1] & n \leq 0. \end{cases}$$

Now it is evident that $y_3[n] = ay_1[n] + by_2[n]$, so the system is linear.

3. (5 marks) Sketch the discrete-time signals with each of the following z-transforms:

(a) $X_1(z) = (1 + 2z)(1 + 3z^{-1})(1 - z^{-1})$ with ROC all z

(b) $X_2(z) = \frac{(1+2z)(1+3z^{-1})}{(1-z^{-1})}$ with ROC $|z| > 1$.

(a) Multiplying out gives

$$\begin{aligned} X(z) &= (1 + 2z)(1 + 2z^{-1} - 3z^{-2}) \\ &= 1 + 2z^{-1} - 3z^{-2} + 2z + 4 - 6z^{-1} \\ &= 2z + 5 - 4z^{-1} - 3z^{-2} \end{aligned}$$

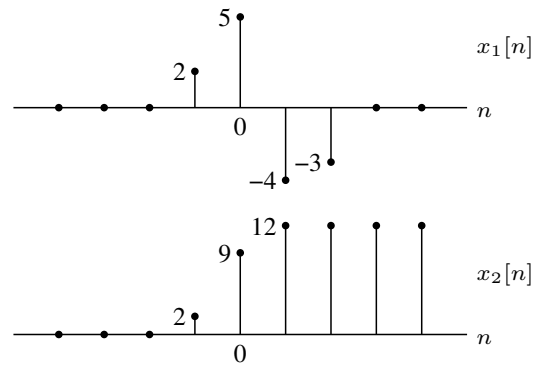
By inspection this inverts to $x_1[n] = 2\delta[n+1] + 5\delta[n] - 4\delta[n-1] - 3\delta[n-2]$.

(b) The given transform evaluates to

$$X_2(z) = \frac{2z + 7 + 3z^{-1}}{(1 - z^{-1})}$$

and the inverse is $x_2[n] = 2u[n+1] + 7u[n] + 3u[n-1]$.

Plots below:



4. (5 marks) We require a casual discrete-time LTI system with the property that if the input is

$$x[n] = (1/2)^n u[n] - 1/4(1/2)^{n-1} u[n-1]$$

then the output is

$$y[n] = (1/3)^n u[n].$$

Determine the impulse response $h[n]$ and the system function $H(z)$ of a system that satisfies these conditions.

The signals in the transform domain are

$$X(z) = \frac{1}{1 - 1/2z^{-1}} - \frac{1/4z^{-1}}{1 - 1/2z^{-1}} = \frac{1 - 1/4z^{-1}}{1 - 1/2z^{-1}}$$

and

$$Y(z) = \frac{1}{1 - 1/3z^{-1}}$$

The system function is therefore

$$\begin{aligned} H(z) &= \frac{Y(z)}{X(z)} = \frac{1 - 1/2z^{-1}}{(1 - 1/3z^{-1})(1 - 1/4z^{-1})} = \frac{z(z - 1/2)}{(z - 1/3)(z - 1/4)} \\ &= \frac{a}{1 - 1/3z^{-1}} + \frac{b}{1 - 1/4z^{-1}} \end{aligned}$$

with $a = -2$ and $b = 3$. The system has poles at $z = 1/3$ and $z = 1/4$. For a causal system the ROC must extend outwards from the outermost pole, so the required ROC is $|z| > 1/3$. The impulse response is therefore

$$h[n] = -2(1/3)^n u[n] + 3(1/4)^n u[n].$$

Transforms

$$\begin{aligned}
 X(j\Omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt & \iff & x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega & \text{(CTFT)} \\
 X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} & \iff & x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega & \text{(DTFT)} \\
 X(z) &= \sum_{n=-\infty}^{\infty} x[n]z^{-n} & \iff & x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz & \text{(ZT)} \\
 X[k] &= \sum_{n=0}^{N-1} x[n]W_N^{kn} & \iff & x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}} & \text{(DFT)}
 \end{aligned}$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)}) d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n + 1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$